

# 1. Introduction

Milestone 2 focuses on performing deeper statistical analysis on the cleaned weather dataset and creating an interactive dashboard to visualize climate patterns. The goal is to understand global distributions, correlations, seasonal behavior, extreme weather events, and regional differences.

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## 2. Statistical Analysis Performed

### 2.1 Distribution Analysis

We explored how temperature, humidity, wind speed, precipitation, and air-quality values are distributed.

This helped identify:

- Normal ranges
- Outliers
- Climate behavior patterns

### 2.2 Correlation Analysis

We examined relationships between climate variables such as:

- Temperature and humidity
- Wind speed and air pressure
- Air-quality metrics like PM2.5, PM10, and NO<sub>2</sub>

This helped identify which factors influence each other.

### 2.3 Seasonal & Temporal Patterns

Using timestamps, month and year were extracted.

This allowed us to identify monthly and seasonal patterns such as:

- Rising temperatures during summer

- Changing humidity and wind trends

## 2.4 Trend Analysis

Time-series plots helped visualize long-term climate behavior.

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## 3. Extreme Weather Identification

Using percentile thresholds, we identified:

- Heat waves
- High rainfall events
- Strong wind events

This helped detect unusual or dangerous weather patterns.

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## 4. Regional Comparison

Countries were compared based on:

- Average temperature
- Humidity
- Wind speed
- Air-quality values

This revealed important regional climate differences across the world.

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## 5. Visualization & Dashboard Design

An interactive dashboard was designed using Dash and Plotly.  
It includes:

- A country selection dropdown
- A temperature trend line chart
- A global temperature choropleth map

The dashboard automatically opens in the browser and allows users to explore global climate patterns.

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## 6. Conclusion

Milestone 2 successfully includes:

- Statistical analysis
- Extreme weather detection
- Regional comparison
- Visualization planning
- An interactive dashboard

This provides a strong analytical foundation for further insights in the next milestone.